

LAB NOC 12

NAT (PAT) + ACL — Internet Simulation

Cisco 888 ISR + Catalyst 2960 | Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware	Cisco Catalyst 2960 + Cisco 888 ISR	Cisco Catalyst 2960 + Cisco 888 ISR
Ticket ID	NOC-LAB-12	NOC-LAB-12
Categoría / Category	NAT PAT + ACL — Salida a Internet simulada	NAT PAT + ACL — Simulated Internet access
Nivel / Level	CCNA — NAT / ACL / IP Services	CCNA — NAT / ACL / IP Services
Red interna / Inside	VLAN 1 — 192.168.10.0/24 — ip nat inside	VLAN 1 — 192.168.10.0/24 — ip nat inside
Red externa / Outside	VLAN 2 — 200.1.1.0/24 — ip nat outside (Internet simulado)	VLAN 2 — 200.1.1.0/24 — ip nat outside (simulated Internet)
Duración / Duration	Aprox. 45 minutos	Approx. 45 minutes
Relevancia NOC	NAT/PAT en borde WAN, ACL como herramienta de control de tráfico	NAT/PAT at WAN edge, ACL as traffic control tool

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Configurar NAT con PAT (overload) en el Cisco 888 ISR para permitir que la red interna 192.168.10.0/24 salga a Internet usando una sola IP pública (200.1.1.1), usando una ACL estándar para identificar el tráfico a traducir, y documentar el troubleshooting de los 3 fallos clásicos de NAT.	Configure NAT with PAT (overload) on the Cisco 888 ISR to allow the 192.168.10.0/24 internal network to access Internet using a single public IP (200.1.1.1), using a standard ACL to identify traffic to translate, and document troubleshooting of the 3 classic NAT failures.

INCIDENT TICKET — NOC-LAB-12

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-12
Title	Usuarios con IP pero sin salida a Internet / Users have IP but no Internet access
Priority	Medium-High — afecta productividad de toda la red interna / affects entire internal network productivity
Symptom	ping 200.1.1.2 (Internet simulado) desde PC falla. PC tiene IP 192.168.10.X correcta. / ping to simulated Internet fails. PC has correct IP 192.168.10.X.
Root Cause	NAT no configurado — tráfico de red privada no se traduce a IP

Requirements	publica / NAT not configured — private network traffic not translated to public IP
	ACL para identificar red interna + ip nat inside/outside en interfaces + ip nat inside source list overload
Resolution	access-list 1 permit 192.168.10.0 0.0.0.255 + ip nat inside source list 1 interface vlan 2 overload
Validation	show ip nat translations → muestra traducciones activas / shows active translations
Status	CERRADO — NAT PAT operativo, ping a Internet simulado exitoso / CLOSED — NAT PAT operational, ping to simulated Internet successful

CONCEPTOS NAT / NAT CONCEPTS

CONCEPTO / CONCEPT	DESCRIPCION (ES)	DESCRIPTION (EN)
Inside Local	IP privada del host en la red interna (192.168.10.10)	Private IP of the host on the internal network (192.168.10.10)
Inside Global	IP publica usada para representar al host en Internet (200.1.1.1)	Public IP used to represent the host on Internet (200.1.1.1)
Outside Global	IP del destino en Internet (200.1.1.2 — simulado)	IP of the Internet destination (200.1.1.2 — simulated)
PAT / Overload	Multiple hosts internos comparten una sola IP publica usando puertos distintos	Multiple internal hosts share a single public IP using different ports
ip nat inside	Marca la interfaz como red interna (VLAN 1 — LAN)	Marks the interface as internal network (VLAN 1 — LAN)
ip nat outside	Marca la interfaz como red externa / Internet (VLAN 2 — WAN)	Marks the interface as external/Internet network (VLAN 2 — WAN)
ACL en NAT	La ACL no bloquea trafico — identifica que trafico sera traducido por NAT	ACL does not block traffic — identifies which traffic will be translated by NAT

DIRECCIONAMIENTO — INSIDE / OUTSIDE

VLAN	ROL NAT	RED / NETWORK	IP ROUTER	IP HOST	USO / USE
VLAN 1	ip nat inside	LAN interna	192.168.10.1 /24	192.168.10.10	PC / dispositivos internos
VLAN 2	ip nat outside	WAN / Internet simulado	200.1.1.1 /24	200.1.1.2	Destino Internet simulado

EJEMPLO DE TRADUCCION NAT / NAT TRANSLATION EXAMPLE

Inside Local	Inside Global	Outside Local	Outside Global
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192.168.10.10:1234	200.1.1.1:1234	200.1.1.2	200.1.1.2
IP privada del host	IP publica del router (NAT)	Destino en Internet	Destino en Internet

PROCEDIMIENTO — 4 ETAPAS / PROCEDURE — 4 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Configurar interfaz LAN (VLAN 1): ip address 192.168.10.1 /24 + ip nat inside	STAGE 1 — Configure LAN interface (VLAN 1): ip address 192.168.10.1 /24 + ip nat inside
2	Configurar interfaz WAN simulada (VLAN 2): ip address 200.1.1.1 /24 + ip nat outside	Configure simulated WAN interface (VLAN 2): ip address 200.1.1.1 /24 + ip nat outside
3	Verificar: show ip interface brief → VLAN 1 y VLAN 2 up/up con IPs correctas	Verify: show ip interface brief → VLAN 1 and VLAN 2 up/up with correct IPs
4	ETAPA 2 — Crear ACL para NAT: access-list 1 permit 192.168.10.0 0.0.0.255	STAGE 2 — Create ACL for NAT: access-list 1 permit 192.168.10.0 0.0.0.255
5	Configurar PAT: ip nat inside source list 1 interface vlan 2 overload	Configure PAT: ip nat inside source list 1 interface vlan 2 overload
6	Configurar default route: ip route 0.0.0.0 0.0.0.0 200.1.1.2	Configure default route: ip route 0.0.0.0 0.0.0.0 200.1.1.2
7	end → write memory	end → write memory
8	ETAPA 3 — Verificación: ping 200.1.1.2 desde PC	STAGE 3 — Verification: ping 200.1.1.2 from PC
9	show ip nat translations → confirmar traducciones activas (Inside Local ↔ Inside Global)	show ip nat translations → confirm active translations (Inside Local ↔ Inside Global)
10	show ip nat statistics → verificar hits y total translations > 0	show ip nat statistics → verify hits and total translations > 0
11	ETAPA 4 — Debug avanzado: debug ip nat + ping → observar traducciones en tiempo real	STAGE 4 — Advanced debug: debug ip nat + ping → observe translations in real time
12	undebg all → cerrar debug. Ticket NOC-LAB-12 cerrado con evidencia.	undebg all → close debug. Ticket NOC-LAB-12 closed with evidence.

LAB NOC 12

NAT + ACL — Simulando Internet

Objetivo

Permitir que una red interna **192.168.10.0/24** salga hacia Internet usando **NAT (PAT)** en el router.

1Internet simulado
200.1.1.0

|
Router Cisco 888

VLAN2 (outside)

IP 200.1.1.1

|

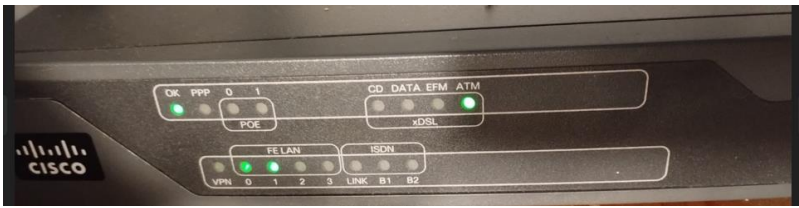
Switch 2960

Fa0/10

PC / Laptop

VLAN1 (inside)

192.168.10.0.



2. Escenario

Un usuario reporta:

"Tengo IP pero no puedo salir a Internet".

El técnico NOC debe verificar **NAT en el router**.

3. Configurar interfaz LAN

En el router:

```
conf t
interface vlan 1
ip address 192.168.10.1 255.255.255.0
ip nat inside
no shutdown
exit
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface vlan 1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#
*Mar  8 23:26:22.260: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#no shut
R1(config-if)#exit
```

4. Configurar interfaz WAN

Ejemplo (FastEthernet1):

```
interface vlan 2
ip address 200.1.1.1 255.255.255.0
ip nat outside
no shutdown
exit
```

```
R1(config-if)#exit
R1(config)#interface vlan 2
R1(config-if)#ip ad
*Mar  8 23:32:14.553: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to
R1(config-if)#ip address 200.1.1.1 255.255.255.0
R1(config-if)#ip nat outside
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fal
R1(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
```

VLAN 1 → red interna (192.168.10.0)

VLAN 2 → red externa simulada (200.1.1.0)

```
COM5 - PuTTY
VLAN mismatch discovered on FastEthernet0/9 (1), with R1 FastEthernet1 (2).
Switch>
Switch>
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#no shut
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
*Mar  1 11:52:35.005: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discover
ed on FastEthernet0/9 (1), with R1 FastEthernet1 (2).
*Mar  1 11:53:30.529: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discover
ed on FastEthernet0/9 (1), with R1 FastEthernet1 (2).
```

Command show ip interface brief en router

```
COM5 - PuTTY
ocol
BRI0                unassigned      YES NVRAM    administratively down down
BRI0:1              unassigned      YES unset    administratively down down
BRI0:2              unassigned      YES unset    administratively down down
FastEthernet0        unassigned      YES unset    up            up
FastEthernet1        unassigned      YES unset    up            up
FastEthernet2        unassigned      YES unset    up            down
FastEthernet3        unassigned      YES unset    up            down
NVI0                 10.10.10.1      YES unset    up            up
Vlan1                 192.168.10.1    YES manual   up            up
Vlan2                 200.1.1.1       YES manual   up            up
Vlan10               unassigned      YES unset    down          down
R1#
```

5. Crear ACL para NAT

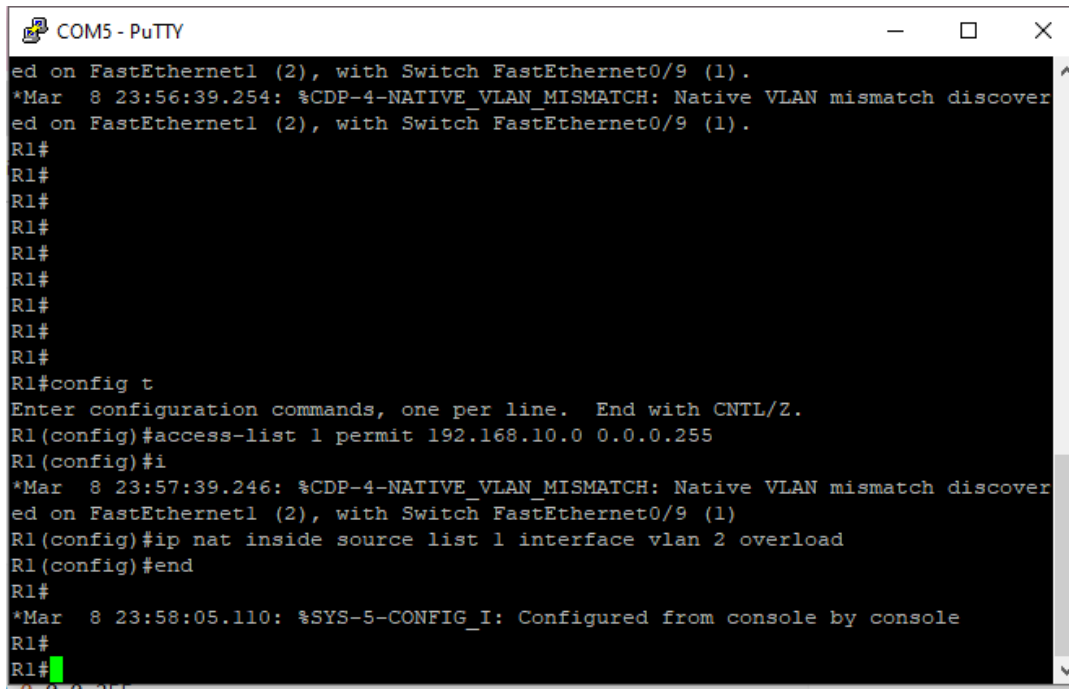
NAT necesita una ACL que identifique la red interna.

```
access-list 1 permit 192.168.10.0 0.0.0.255
```

6. Configurar NAT overload (PAT)

```
ip nat inside source list 1 interface fa1 overload
```

Esto permite que **toda la red interna salga con una sola IP pública.**

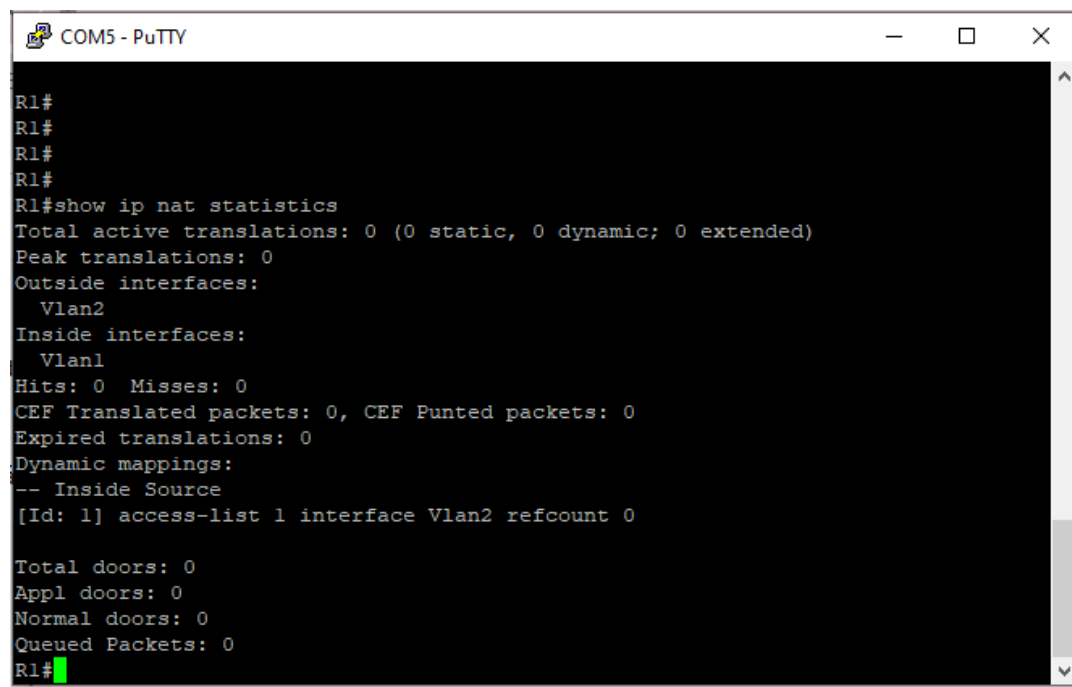


```
ed on FastEthernet1 (2), with Switch FastEthernet0/9 (1).
*Mar  8 23:56:39.254: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discover
ed on FastEthernet1 (2), with Switch FastEthernet0/9 (1).
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#i
*Mar  8 23:57:39.246: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discover
ed on FastEthernet1 (2), with Switch FastEthernet0/9 (1)
R1(config)#ip nat inside source list 1 interface vlan 2 overload
R1(config)#end
R1#
*Mar  8 23:58:05.110: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
```

7. Verificar NAT

Comandos clave de NOC:

show ip nat translations
show ip nat statistics
show running-config | include nat



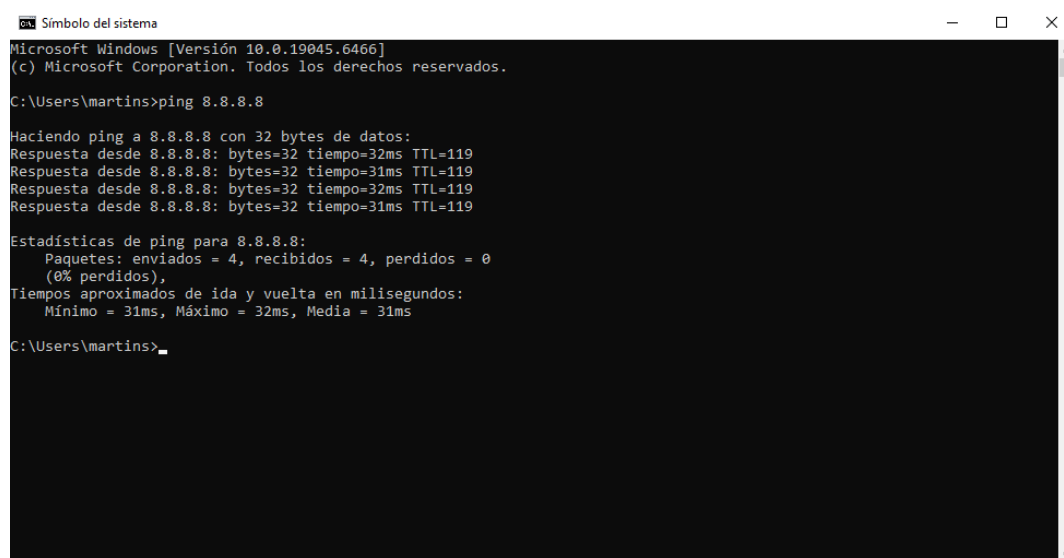
```
COM5 - PuTTY
R1#
R1#
R1#
R1#
R1#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Vlan2
Inside interfaces:
  Vlan1
Hits: 0 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 interface Vlan2 refcount 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1#
```

8. Probar conectividad

Desde la PC:

ping 8.8.8.8



```
Símbolo del sistema
Microsoft Windows [Versión 10.0.19045.6466]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\martins>ping 8.8.8.8

Haciendo ping a 8.8.8.8 con 32 bytes de datos:
Respuesta desde 8.8.8.8: bytes=32 tiempo=32ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=31ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=32ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=31ms TTL=119

Estadísticas de ping para 8.8.8.8:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
              (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 31ms, Máximo = 32ms, Media = 31ms

C:\Users\martins>
```

9. Ver NAT funcionando

En el router debería aparecer algo como:

show ip nat translations


```
R1#
R1#
R1#show running-config | include nat
  isdn termination multidrop
  ip nat inside
  ip nat outside
ip nat inside source list 1 interface Vlan2 overload
R1#
```

Ejemplo:

Inside global	Inside local
200.10.10.5	192.168.10.10

10. Troubleshooting típico NOC

Si NAT no funciona revisar:

1 ACL incorrecta

show access-lists

2 Interfaces NAT mal definidas

show running-config | include nat

Debe aparecer:

```
ip nat inside
ip nat outside
```

3No hay traducciones

show ip nat translations

Si está vacío → NAT no está funcionando.

11. Debug NAT (avanzado)

debug ip nat

Luego probar ping.

Para apagar:

undebug all

Conclusión del incidente NOC

Problema reportado:

Usuarios no pueden salir a Internet.

Diagnóstico:

Configuración NAT incorrecta o inexistente.

Resolución:

Se configuró **PAT (NAT overload)** en el router.

Resultado:

Usuarios pueden acceder a Internet.

CONFIGURACION PASO A PASO / STEP-BY-STEP CONFIGURATION

#	COMANDO / COMMAND	ACCION (ES)	ACTION (EN)
1	<code>conf t</code>	Modo configuracion global	Global configuration mode
2	<code>interface vlan 1 → ip address 192.168.10.1 255.255.255.0 → ip nat inside → no shutdown → exit</code>	Configurar interfaz LAN con rol NAT inside	Configure LAN interface with NAT inside role
3	<code>interface vlan 2 → ip address 200.1.1.1 255.255.255.0 → ip nat outside → no shutdown → exit</code>	Configurar interfaz WAN con rol NAT outside	Configure WAN interface with NAT outside role
4	<code>access-list 1 permit 192.168.10.0 0.0.0.255</code>	Crear ACL que identifica la red interna para NAT	Create ACL that identifies internal network for NAT
5	<code>ip nat inside source list 1 interface vlan 2 overload</code>	Configurar PAT: toda la red interna sale con IP de VLAN 2	Configure PAT: entire internal network exits with VLAN 2 IP
6	<code>ip route 0.0.0.0 0.0.0.0 200.1.1.2</code>	Default route hacia Internet simulado (necesaria para routing)	Default route toward simulated Internet (required for routing)
7	<code>end → write memory</code>	Salir y persistir toda la configuracion NAT	Exit and persist all NAT configuration

TROUBLESHOOTING NAT — 3 FALLAS CLASICAS / 3 CLASSIC NAT FAILURES

#	FALLA / FAILURE	DIAGNOSTICO / DIAGNOSIS	SINTOMA / SYMPTOM	RESOLUCION / RESOLUTION
1	ACL incorrecta	<code>show access-lists</code>	ACL 1 no matchea trafico de 192.168.10.0 — wildcard mask erronea o red equivocada / ACL 1 not matching traffic from 192.168.10.0 — wrong wildcard mask or wrong network	Recrear ACL: access-list 1 permit 192.168.10.0 0.0.0.255. Nota: 0.0.0.255 es wildcard, no mascara de subred / Recreate ACL. Note: 0.0.0.255 is wildcard, not subnet mask
2	Interfaces NAT mal definidas	<code>show running-config include nat</code>	Una interfaz sin ip nat inside o ip nat outside — el router no sabe que interfaz es LAN y	Aplicar ip nat inside en VLAN 1 y ip nat outside en VLAN 2. Ambas deben estar

3	Sin traducciones activas	show ip nat translations	cual es WAN / One interface missing ip nat inside or ip nat outside	configuradas / Apply ip nat inside on VLAN 1 and ip nat outside on VLAN 2. Both must be configured
			Tabla NAT vacia aunque el ping se envio — trafico no esta siendo traducido / Empty NAT table even though ping was sent	Verificar en orden: ACL → interfaces NAT → default route. Usar debug ip nat + ping para ver en tiempo real / Verify in order: ACL → NAT interfaces → default route. Use debug ip nat + ping to see in real time

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
show ip nat translations	Muestra tabla de traducciones activas: Inside Local ↔ Inside Global	Shows active translation table: Inside Local ↔ Inside Global
show ip nat statistics	Estadísticas NAT: hits, misses, traducciones totales	NAT statistics: hits, misses, total translations
show running-config include nat	Verifica rapido si ip nat inside/outside y overload estan configurados	Quick check if ip nat inside/outside and overload are configured
show access-lists	Muestra ACLs y su contador de matches — util para verificar que la ACL matchea trafico	Shows ACLs and their match counters — useful to verify ACL is matching traffic
debug ip nat	Muestra traducciones NAT en tiempo real — usar con cuidado en produccion	Shows NAT translations in real time — use carefully in production
undebug all	Detiene todos los debugs activos en el router	Stops all active debugs on the router
clear ip nat translation *	Limpia todas las traducciones NAT activas	Clears all active NAT translations

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
La ACL para NAT usa mascara de subred en vez de wildcard mask. access-list 1 permit 192.168.10.0 255.255.255.0 NO funciona para NAT. / ACL for NAT uses subnet mask instead of wildcard mask. access-list 1 permit 192.168.10.0 255.255.255.0 does NOT work for NAT.	La wildcard mask es el inverso de la mascara: /24 = 0.0.0.255. Usar siempre access-list 1 permit 192.168.10.0 0.0.0.255. Verificar con show access-lists que el match counter sube con el trafico. / Wildcard mask is the inverse of subnet mask: /24 = 0.0.0.255. Always use access-list 1 permit 192.168.10.0 0.0.0.255. Verify with show access-lists that match counter increases with traffic.
El comando ip nat inside source apunta a la interfaz fisica 'fa1' pero en el 888 con SVIs se	En el Cisco 888 con SVIs: ip nat inside source list 1 interface vlan 2 overload. Si se usa una interfaz

debe apuntar a 'vlan 2'. / The ip nat inside source command points to physical interface 'fa1' but on the 888 with SVIs it must point to 'vlan 2'.

show ip nat translations esta vacio aunque la config parece correcta — NAT no esta traduciendo. / show ip nat translations is empty even though config looks correct — NAT is not translating.

fisica directa seria 'interface fastEthernet1'. Verificar siempre con show running-config | include nat que la interfaz es correcta. / On Cisco 888 with SVIs: ip nat inside source list 1 interface vlan 2 overload. Verify with show running-config | include nat that interface is correct.

Verificar en orden: (1) default route presente (show ip route), (2) ACL matcheando trafico (show access-lists — contador debe subir), (3) debug ip nat + ping para ver el flujo en tiempo real. Si debug no muestra nada, el problema es antes de NAT (L2/L3). / Verify in order: (1) default route present (show ip route), (2) ACL matching traffic (show access-lists — counter must increase), (3) debug ip nat + ping to see flow in real time.

RESULTADO / RESULT

RESULTADO (ES)

NAT PAT configurado y operativo. Red interna 192.168.10.0/24 sale a Internet simulado (200.1.1.0/24) con IP 200.1.1.1. show ip nat translations confirma traducciones activas. debug ip nat valida flujo en tiempo real. Ticket NOC-LAB-12 cerrado.

RESULT (EN)

NAT PAT configured and operational. Internal network 192.168.10.0/24 exits to simulated Internet (200.1.1.0/24) with IP 200.1.1.1. show ip nat translations confirms active translations. debug ip nat validates flow in real time. Ticket NOC-LAB-12 closed.

KEY TAKEAWAY

"NAT PAT tiene 3 requisitos obligatorios: (1) ACL con wildcard mask correcta — no subnet mask, (2) ip nat inside en la interfaz LAN y ip nat outside en la interfaz WAN — ambas, (3) ip nat inside source list con overload apuntando a la interfaz WAN correcta. Si el show ip nat translations esta vacio, la causa es siempre una de estas tres. El debug ip nat es la herramienta definitiva para confirmar donde falla la traduccion."

"NAT PAT has 3 mandatory requirements: (1) ACL with correct wildcard mask — not subnet mask, (2) ip nat inside on LAN interface and ip nat outside on WAN interface — both required, (3) ip nat inside source list with overload pointing to the correct WAN interface. If show ip nat translations is empty, the cause is always one of these three. debug ip nat is the definitive tool to confirm where translation is failing."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), NAT PAT is the standard mechanism at every remote site router where a private LAN (offices, control rooms) connects to the WAN (corporate MPLS or satellite link). A NOC engineer must diagnose NAT failures quickly because they affect all users simultaneously — unlike single-port issues. The three-command diagnosis sequence (show ip nat translations → show access-lists → show running-config | include nat) resolves 90% of NAT tickets. debug ip nat is reserved for production use only when the other three commands provide no answer, due to its CPU impact on the router.